

Sara Nguyen

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EDUCATION

University of Colorado Boulder

Bachelor of Science in Computer Science

Boulder, CO

Expected May 2028

- **GPA: 3.979/4.0**
- **Relevant Coursework:** Data Structures, Algorithms, Computer Systems, Software Development, Operating Systems, Data Science, Cybersecurity, Database Systems
- **Organizations:** Society of Women Engineers, Society of Asian Scientists and Engineers
- **Certifications:** IBM AI Fundamentals

TECHNICAL SKILLS

- **Languages:** C, C++ (500+ hours), Python (300+ hours), Java, JavaScript, TypeScript, HTML/CSS, SQL
- **Frameworks & Tools:** React.js, Node.js, Express, Docker, Git, GitHub, PostgreSQL, Pandas, Scikit-learn, NumPy

PROJECTS

Sarandipity Booth – Serverless Cloud Photobooth Web App

Aug 2026 – Present

React, TypeScript, Vite, AWS (Cognito, S3, DynamoDB), AWS CDK, Vercel

- Building a browser-based photobooth in React/TypeScript, using the Canvas API to apply live filters and composite multi-shot photo strips entirely client-side.
- Provisioning a serverless AWS backend with CDK (Cognito, S3, DynamoDB), using IAM policies to scope each user to their own data with no API server required.
- Implementing Cognito Identity Pool credential exchange, letting the browser call AWS directly via short-lived, per-user-scoped credentials.

DNS Resolver with Thread-Safe Array

March 2026

Linux, C, POSIX Threads (pthreads), Multithreading, Synchronization, Network Programming, Sockets

- Designed a multi-threaded producer-consumer system in C, with requester threads parsing hostnames and resolver threads concurrently resolving them via getaddrinfo() and inet_ntop().
- Built a thread-safe circular queue using mutexes and condition variables, allowing resolver threads to block on empty and wake efficiently instead of busy-waiting.
- Fixed a deadlock where resolver threads hung after producers finished, using a shared "done" flag with pthread_cond_broadcast() for clean shutdown.

File Tester and Character Device Driver

Feb 2026

Linux, C, Kernel Modules, Device Drivers, POSIX System Calls, Systems Programming

- Built a custom character device driver as a loadable Linux kernel module (LKM), implementing open, read, write, and seek operations at the kernel level.
- Developed a user-space CLI program to interactively issue read, write, and seek commands against the driver, testing correctness against expected file semantics.
- Gained hands-on experience with Linux file descriptors, the kernel's file descriptor table, and low-level POSIX system calls (open, read, write, lseek) bridging user and kernel space.

Buck-It

Oct – Dec 2025

HTML, CSS, JavaScript, Bootstrap, Node.js, Express, PostgreSQL, Render

- Developed a full-stack travel planning application in a five-person Agile team.
- Designed and implemented RESTful backend services to manage user authentication, trips, itineraries, and social interactions.
- Implemented GPS metadata verification with Exifr to authenticate archived travel images and improve trip-sharing reliability.
- Built responsive frontend interfaces for managing destinations, trip statuses, and personalized travel planning features; deployed the application using Render.

WORK EXPERIENCE

Volleyball Coach

Platte River Academy

Aug 2023 – Oct 2023

Highlands Ranch, CO

- Led team to regional championships through strategic game planning and skill development.
- Designed and ran efficient practice sessions focused on teamwork, technique, and sportsmanship.
- Fostered teamwork and leadership.

LEADERSHIP

- **Vietnamese Student Association:** Executive Board Intern (2025–2026); current Culture Show Coordinator leading planning for the organization's largest annual event with 600+ attendees.